



Experimental investigation of air-fuel equivalence ratio effects on advanced dual-fuel ammonia/diesel combustion on a single-cylinder medium-duty diesel engine at high load

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ABSTRACT

Ammonia (NH₃) has garnered significant interest as an alternative fuel for meeting international emissions reduction mandates in sectors with high weight and distance requirements, such as shipping. Technical barriers and unanswered questions remain on the combustion strategies that can maximize ammonia utilization and minimize emissions. Prior research studies at the US Department of Energy's Oak Ridge National Laboratory have shown strong performance with NH₃ under dual-fuel mode using conventional diesel combustion (CDC) manifold air pressure settings. Diesel airflow was initially used to simplify retrofitting (no turbocharger modification), which resulted in air-fuel equivalence ratios (λ) greater than 1.5. To characterize potential improvements in dual-fuel NH₃ combustion performance at richer in-cylinder conditions, a global λ sweep compared the use of early (E-pilot) and late (L-pilot) single diesel injections. The experiments were conducted at 1200 RPM and 12.8 ± 0.2 bar (75 % load), and λ was varied by decreasing the commanded air flow to the engine at greater than 90 % ammonia energy substitution level. A diesel injection timing sweep was conducted for both the injection strategies at fixed λ , and the timing with the lowest engine-out N₂O emissions was identified. The results indicated an optimal balance between CO_{2,eq} and thermal efficiency benefits both E-pilot and L-pilot injection strategy cases compared with CDC at a λ of 1.4. The indicated nitrogen-based emissions exhibited a strong correlation to the ratio of CA₅₋₅₀ and ignition delay for L-pilot, but no apparent trend emerged for the E-pilot injection strategy at the tested boundary conditions.

1. Introduction

Maritime transportation is a significant source of global air pollutant emissions, and there has been increasing attention in recent years to reducing these emissions. Energy utilization—and thus emissions—for international shipping is projected to increase 50 %–250 % between now and 2050, subject to economic conditions [1]. The International Maritime Organization (IMO), meanwhile, has committed to reduce CO₂-equivalent emissions from ships by 70 %–80 % from 2008 levels by 2040 [1,2], resulting in the need for consideration of alternative fuels.

Ammonia (NH₃) has garnered significant interest as a new fuel for maritime applications to power freighters and tankers. It is the second-most produced chemical in the world (~200 Mt/year) via the Haber-Bosch process [3–5] and is widely shipped, with significant existing port infrastructure. Its relatively low storage pressure (0.99 MPa vs. 69 MPa for compressed H₂) makes it economical for long-distance

transportation (by comparison, H₂ storage is about 30× more expensive) [6,7].

Ammonia's use as a fuel, however, comes with a unique set of challenges. The apparent toxicity (vapor pressure relative to toxicity at atmospheric temperature) of NH₃ is approximately three orders of magnitude greater than that of methanol or gasoline [8]. Its cetane number, lower than that of diesel (~0 for NH₃ vs. >40 for diesel), makes its use as a compression ignition (CI) fuel difficult. Furthermore, its low laminar flame speed (gasoline is about 6x higher at standard conditions) and high ignition energy requirement (~2000× higher than gasoline at standard conditions) [9,10] make it challenging to burn in a spark-ignited (SI) application. The combustion management of NH₃ is crucial because its partial oxidation forms nitrous oxide (N₂O), which has a multiplier of 273 for CO₂-equivalent emissions over a span of 100 years [11,12]. High engine-out NO_x due to fuel-bound nitrogen and unburned NH₃ emissions are also issues that require attention when

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using NH_3 as a fuel.

Considerable effort has been invested in alternative injection strategies that use a combination of low- and high-ignition-quality fuels to lower CO_2 -equivalent emissions. Strategies such as diesel pilot ignition (DPI), which uses a late diesel injection (near-top dead center (TDC)), and reactivity-controlled compression ignition (RCCI), which uses an early diesel injection, enable the use of new fuels like NH_3 in a compression ignition engine. In the DPI strategy, fuels like diesel are used as the high-reactivity pilot to ignite a low-reactivity premixed charge (NH_3/air in this instance). Typically, as seen in DPI combustion of methane/air mixtures, the pilot fuel creates multiple ignition sites that result in flame fronts that propagate through the rest of the premixed charge, resulting in a double-humped heat release rate. This strategy allows for high-load operation with reasonable pressure rise rates but produces more NO_x and particulate matter (PM) emissions than strategies employing earlier diesel injections like RCCI [13–16]. The RCCI strategy results in a rapid volumetrically cascading combustion progression due to in-cylinder fuel reactivity stratification caused by the early injection of the high-reactivity fuel (diesel), which helps to control the combustion phasing and duration. However, it is prone to load limitations from high in-cylinder pressure rise rates as well as combustion control issues at low loads with fuels like gasoline and methane [17, 18].

Existing literature on dual-fuel NH_3 combustion covers a range of approaches used by different researchers [19–23]. For instance, studies referenced in Yousefi et al. [19], Liu and Liu [20], and Zheng et al. [21] primarily evaluate the influence of various control parameters on diesel/ NH_3 dual-fuel combustion at ammonia energy substitution (AES) levels of <70 %. Work by Nadimi et al. [22] also utilized near-TDC diesel timings in DPI to achieve high-AES operation (~85 %); however, the effect of λ was not studied in their work. A recent study published by Hiraoka et al. [23] included a comprehensive experimental and numerical analysis of dual-fuel NH_3 at AES as high as 95 %. However, the diesel start of injection (SOI) timings were varied to maintain constant combustion phasing (CA50), and the effects of λ were evaluated at 57 % AES in their approach. The present study contributes to the literature by evaluating the influence of air-fuel equivalence ratio (λ) at AES greater than 90 %, while simultaneously comparing the dual-fuel performance of the early (E-pilot) and late (L-pilot) single diesel injection strategies.

2. Methods/experimental

2.1. Experimental setup and fuels

The platform used for the present study was a single-cylinder version of a Cummins 6.7 L ISB diesel engine. The combustion chamber geometry was not modified from the stock configuration. The stock high-pressure common rail direct-injection fueling system and pump were used for the #2 diesel fuel (Haltermann Solutions, product code HF0582) along with a CRIN-3 eight-hole injector with a 145° included angle and a nominal diameter of 140 μm . The premixed fuel, chemical grade anhydrous NH_3 (Airgas), was drawn from a bottle (50 lb. capacity) with a liquid dip tube and vaporized using a vaporizer before being measured (Emerson Micro Motion Coriolis flowmeter) and injected (two Clean Air Power DigiJet NH3ICE SP-010 injectors) downstream of the intake surge tank (approximately 35 cm upstream of the intake port, in a y-section to allow for mixing with the intake air). No port-fuel injection spray targeting optimization was conducted for this work. Hot engine out coolant was used to maintain the supply of ammonia above its saturation temperature at bottle pressures ranging from 6–8 bar, and all the supply lines and flowmeter were insulated with heat tape to prevent phase change. All fuel system components were evaluated for NH_3 compatibility prior to operation. The vaporized fueling methodology prior to injection effectively removed the heat of vaporization effects of NH_3 from the combustion process, leaving only the chemical effects. The NH_3 supply setup was interlocked with cell ventilation and gas monitors

for leak detection to ensure safe operation. The crankcase was also ventilated to account for NH_3 accumulation. These measures were based on an internal hazard and operability study that was completed for the use of NH_3 in a laboratory system. The compressed air for the intake charge was conditioned using a series of air dryers and particulate filters. The air temperature was controlled using a 6 kW air heater, and the mass flow rate was measured using a mass flow controller from Alicat Scientific. Relevant nitrogen-based (N-based) emissions were measured using a Fourier transform infrared spectrometer (MKS FTIR) along with California Analytical Instrument analyzers for exhaust O_2 (paramagnetic) and unburned hydrocarbon (flame ionization detector) measurements. The exhaust sample lines were heated (to 190 $^\circ\text{C}$) to avoid condensation of water in the exhaust and the FTIR measurement method for NH_3 was obtained from the supplier.

The engine specifications are shown in Table 1, and the fuel properties are shown in Table 2. Curran et al. [24,25] provide additional details relevant to the test setup. The engine was controlled using an open LabVIEW-based engine control system.

2.2. Experimental operating conditions

This study focused on examining improvements that potentially could be offered by richer in-cylinder conditions (lower λ) for dual-fuel diesel/ NH_3 combustion at greater than 90 % AES. The two injection strategies, E-pilot and L-pilot, were implemented by varying the diesel SOI timing at a fixed air-fuel equivalence ratio. The air-fuel equivalence ratio for each mode was swept by varying the commanded airflow (i.e., varying the intake pressure (P_{Intake})) to the engine, while maintaining an AES level between 90 % and 96 % (or a diesel energy share of 10 % to 4 %). For a given injection strategy, an optimal SOI timing based on the lowest engine-out N_2O emissions was determined at a fixed λ value. This was performed to determine the maximum attainable CO_2 -equivalent emissions reduction relative to CDC, which was a primary concern in this work. CDC operation (pilot + main) at baseline manifold air pressure settings and matched engine load was used to provide a benchmark against which the two injection strategies could be evaluated.

The operating conditions for the experiments were as follows: engine speed of 1200 RPM; engine load of 12.8 ± 0.2 bar (75 %); intake temperature of 60 $^\circ\text{C}$; oil and coolant temperatures of 95 $^\circ\text{C}$; $P_{\text{Intake}} - P_{\text{Exhaust}}$ (exhaust pressure) of ~15 kPa; and start of injection for NH_3 at 320° bTDC. The injection pressure of the diesel was ~1000 bar for the L-pilot cases and ~1200 bar for the E-pilot cases. The design of experiments is shown in Table 3. A total of 500 engine cycles were acquired; and the low-speed data were collected simultaneously.

3. Results and discussion

3.1. In-cylinder pressure and apparent heat release rate

The filtered in-cylinder pressure data for the two injection strategies are shown in Fig. 1. The trend in the motored pressure is explained by a decrease in the trapped air mass at lower λ values (richer conditions). The L-pilot cases exhibit an increase in the peak in-cylinder pressure at richer conditions (from 110 bar at $\lambda = 1.6$ to 130 bar at $\lambda = 1.0$), whereas the E-pilot cases, in general, result in an opposite trend (from 151 bar at

Table 1
Engine specifications.

	Value
Connecting Rod Length [mm]	191.9
Bore $\times$ Stroke [mm]	107 $\times$ 124
Displacement (1-cylinder) [L]	1.12
Compression Ratio	20:1
Direct Injection System	High pressure common-rail pump (Stock)
Port Injection System	Ammonia cylinders ($\times 3$)/vaporizer

Table 2

Properties of the direct-injected and premixed fuels (1 bar and 20 °C).

Properties	#2 Diesel	Ammonia
CN	40.8	N/A
C [wt %]	86.8	N/A
H [wt %]	13.2	17.76
N [wt %]	N/A	82.24
Density [g/cc]	0.856	0.609
LHV [MJ/kg]	42.24	18.8

Table 3

Design of experiments.

Mode	Global λ	AES [%]	Optimal SOI Timing [°bTDC]
L-pilot	1.0	94 ± 0.1	9
	1.1	94 ± 0.1	6
	1.4	96 ± 0.2	6
	1.6	96 ± 0.1	3
E-pilot	1.0	90* ± 0.3	24
	1.1	91* ± 0.1	26
	1.4	94 ± 0.2	25
	1.6	95 ± 0.1	24

Note: Single diesel injection used for L-pilot and E-pilot.

* Maximum AES level resulting in detectable combustion.

$\lambda = 1.6$ to 90 bar at $\lambda = 1.0$). It should be noted that a λ of 1.1 for E-pilot does have a lower peak pressure than λ of 1.0, although the combustion is highly unstable in this operation window (COV_{IMEPn} of 4.4 % and 5.8 % at λ of 1.0 and 1.1, respectively).

The apparent heat release rate (AHRR) results shown in Fig. 1 for the two injection strategies exhibit different behaviors as a function of λ . The

L-pilot strategy with near-TDC diesel timings does not show signs of low-temperature heat release (LTHR) or diesel decomposition in the bulk premixed phase, as evident by the zoomed-in window shown on the AHRR plot (left) in Fig. 1. The late injection of the diesel fuel into a high pressure and temperature environment does not allow enough time for the LTHR to distinctly develop, culminating directly into the main heat release. The AHRRs for the richer cases (λ of 1.0 and 1.1) appear volumetric in nature without a slow tail near the end of combustion, whereas the leaner conditions display a prolonged tail. This may be explained by the penetration length (L) estimates for the L-pilot cases shown in Fig. 2 (left) as a function of the time difference from SOI normalized by the ignition time. The penetration length results are normalized by the projected normal distance from the injector tip to the cylinder liner along the spray axis (L_{max}) to avoid overlapping of results at different λ values, and the calculations were performed as outlined in Hiroyasu and Arai [26]. At the richer conditions, the diesel jet travels further into the combustion chamber, especially in the L-pilot strategy, owing to the lower in-cylinder ambient pressure and thus has access to more NH₃ in the peripheral region. This, along with longer mixing times due to lower SOI temperatures (see Fig. 2 middle) causes the most reactive local zones to have the ideal equivalence ratio for volumetric combustion. The prolonged tail seen for the leaner cases is perhaps an artifact of smaller L/L_{max} , which may result in a larger volume of the premixed NH₃/air mixture with minimal-to-no diesel before ignition. The portion with sufficient diesel may autoignite, whereas the rest appears to rely on weak flame propagation, resulting in a tail near the end of combustion.

On the other hand, the E-pilot cases exhibit LTHR behavior across all λ values, as evident from the zoomed-in window shown on the AHRR plot (right) in Fig. 1. The low pressure and temperature environment (see Fig. 2 middle) experienced by the diesel provides sufficient time for

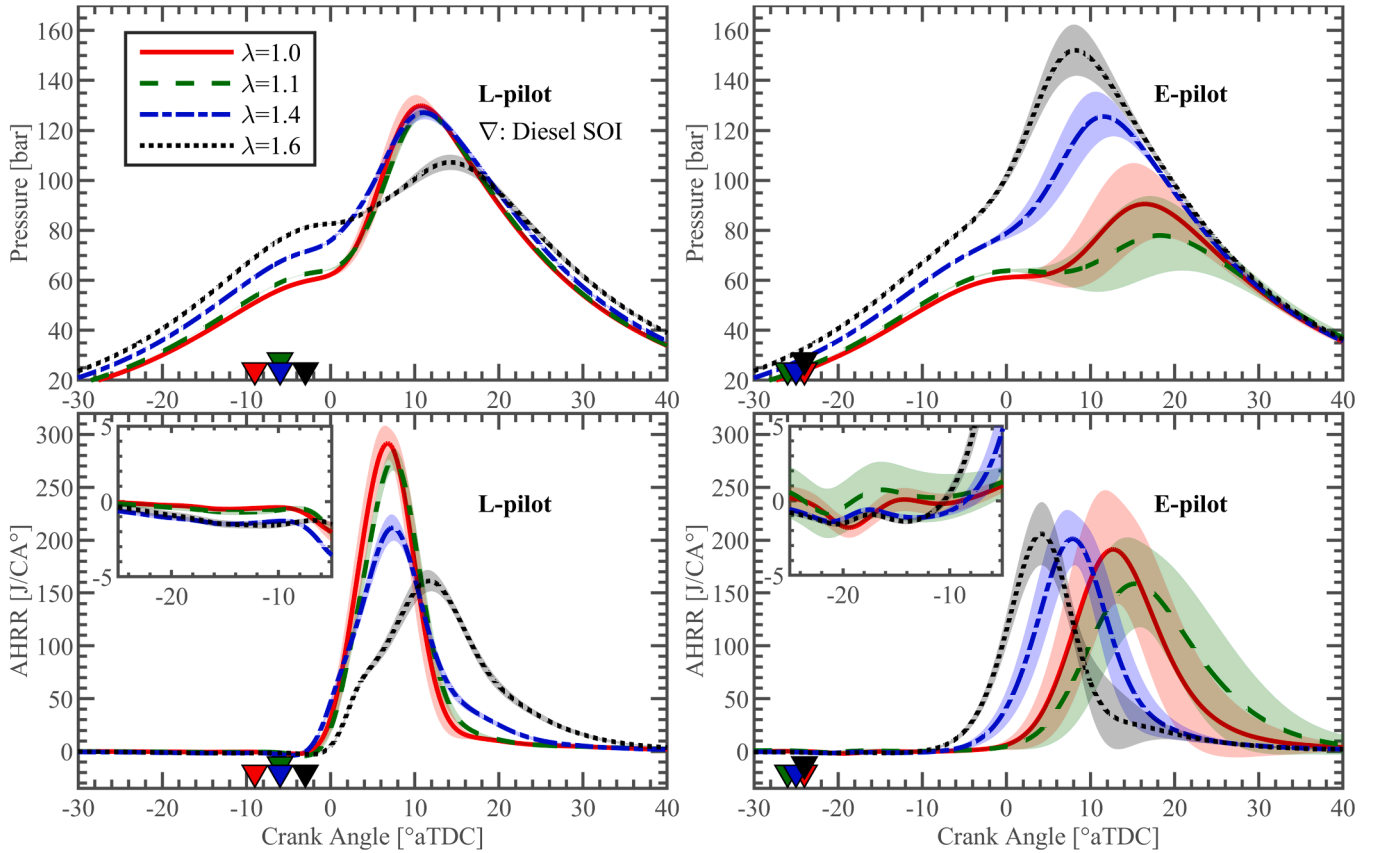


Fig. 1. In-cylinder pressure and apparent heat release rate (AHRR) results at the tested global air-fuel equivalence ratios for the L-pilot and E-pilot injection strategies. Shaded bands represent one standard deviation about the mean. The commanded start of diesel injection is denoted by filled triangular symbols for the different λ cases.

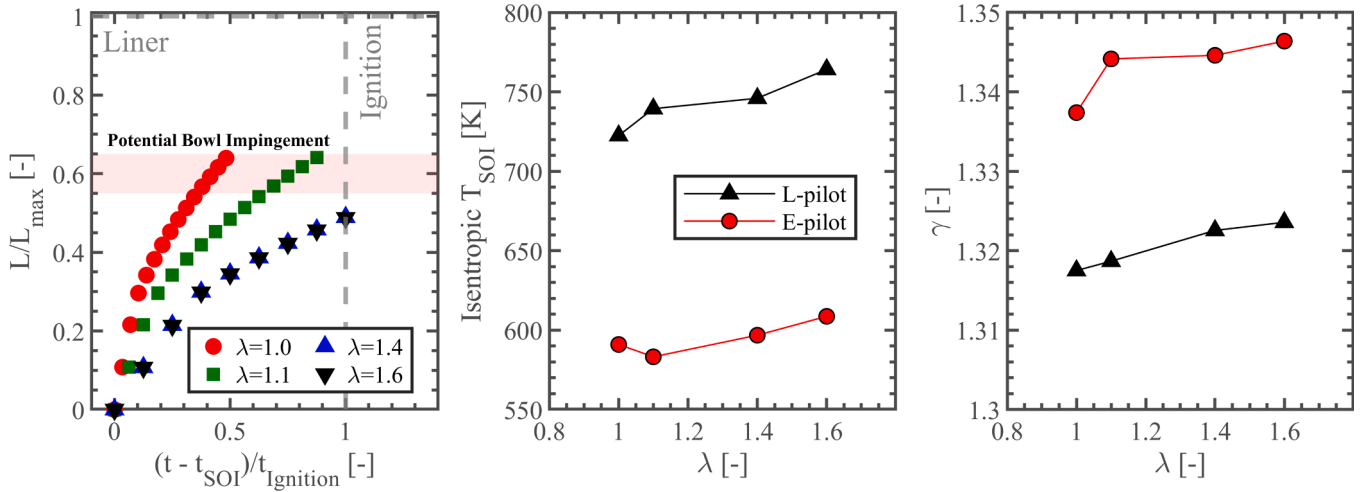


Fig. 2. The ratio of diesel spray penetration length to liner distance along the spray axis as a function of the ratio of time difference from SOI to ignition time is shown for the l-pilot cases (left). Isentropic estimates of SOI temperature (middle) and specific heat ratio (γ ; right) from Cantera are shown for the l-pilot and E-pilot injection strategies.

the LTHR to develop as a small yet noticeable peak in the bulk premixed phase. The LTHR, as noted by Szybist and Splitter [27] for SI combustion of fuels like isooctane, appears as a distinct peak for cases with higher temperature (λ of 1.4 and 1.6), whereas it progresses through the main heat release at lower temperatures. The higher-pressure conditions (λ of 1.4 and 1.6) also result in lower peak LTHR, as evident from Fig. 1. For all the E-pilot cases, the early injection allows for at least adequate mixing of the diesel with the surrounding premixed charge. Considering this, the AHRR for the leaner conditions (λ of 1.4 and 1.6) appears more volumetric in nature. This does not seem to be the case for the richer conditions, however, as suggested by a rather gradually developing main heat release. The lower compression temperatures at SOI, as a result of reduced specific heat ratio (see Fig. 2 right), along with sub-optimal equivalence ratio of the local diesel reaction zones, can impede the autoignition process of the diesel. Hence, the impact of λ on stability, as denoted by the larger standard deviation bands in Fig. 1, is more profound than that in an l-pilot scenario. The authors postulate that the combustion process for this operation strategy is primarily of a lean-premixed reaction propagation type; whether this consists of flamelets or successively cascading autoignition pockets or is driven by deflagration is currently unknown.

3.2. CO₂ emissions and thermal efficiency

The total indicated specific CO₂-equivalent emissions (expressed as CO_{2,eq} emissions, calculated as [ISCO₂] + 273 [ISN₂O]) and thermal efficiency values are overlaid with the baseline CDC results in Fig. 3. The comparison is important to ensure that the CO_{2,eq} reduction with NH₃ is not accompanied by significant thermal efficiency penalties.

The results indicate that dual-fuel NH₃ combustion leads to significant CO_{2,eq} benefits: a reduction greater than 75 % compared with CDC at the studied experimental conditions. Diesel-like air flow ($\lambda = 1.6$) results in the highest CO_{2,eq} penalty for both the l-pilot and E-pilot modes, driven by the incomplete combustion of the premixed fuel, which forms undesirable N₂O at leaner conditions. A change in λ from 1.6 to 1.4 reveals a substantial percent reduction in CO_{2,eq} emissions for both l-pilot (50 %) and E-pilot (35 %) because of lower N₂O in the exhaust. The l-pilot cases display diminishing returns as λ further decreases, whereas a noticeable increase in CO_{2,eq} is observed for E-pilot (a larger contribution from CO₂ resulting from higher diesel energy share required to sustain combustion; see Table 3). A maximum thermal efficiency of 49 % is also obtained for the l-pilot case with a λ of 1.4, representing an increase of 1 % (absolute) with respect to CDC. A λ of 1.6 for E-pilot results in the highest thermal efficiency increase relative to CDC (1.8 % absolute). However, E-pilot operation at a λ of 1.4 provides

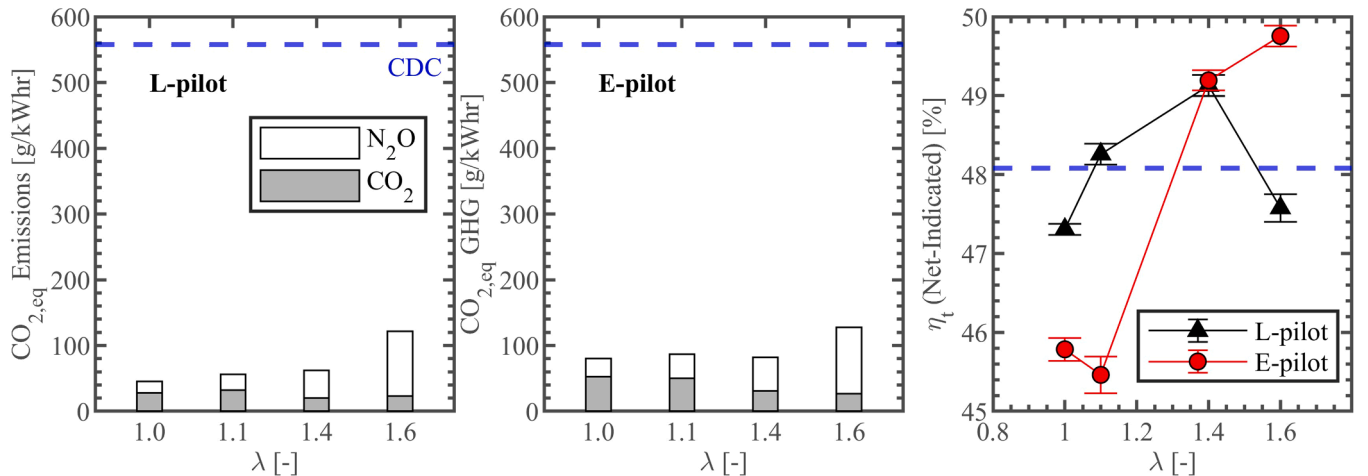


Fig. 3. Engine-out CO_{2,eq} and thermal efficiency results for l-pilot and E-pilot injection strategies compared against baseline diesel-only values. The CO_{2,eq} contribution from N₂O is computed as $273 \times [\text{ISN}_2\text{O}]$ [g/kWhr]. Error bars represent the 95 % confidence interval on the mean.

both the lowest engine-out $\text{CO}_2\text{-eq}$ emissions and a thermal efficiency improvement over CDC.

3.3. N-based emissions

Fig. 4 shows the engine-out NH_3 , NO_x , and N_2O emissions as a function of λ for the two injection strategies. The unburned NH_3 emissions are greater than the NO_x produced in both instances. This reveals a potential opportunity for NO_x conversion over a selective catalytic reduction system using the excess NH_3 without requiring the need for an additional reductant. However, a NH_3/NO_x ratio of 1 is desirable to avoid NH_3 slip over the catalyst, which may necessitate an optimization of $\text{CO}_2\text{-eq}$ and AES. The NO_x emissions peak at a λ of 1.4, whereas the NH_3 emissions converge to a value of ~ 6500 ppm as λ is reduced for l-pilot and E-pilot. The inverse relationship between unburned NH_3 and NO_x suggests that much of the NO_x produced in this instance is fuel-borne and can be thought of as an incomplete combustion product of the premixed NH_3 . The E-pilot strategy, in general, also results in lower unburned NH_3 than that of l-pilot, possibly influenced by the increased spatial availability of the high-reactivity fuel resulting in better consumption of the surrounding premixed NH_3 . Higher unburned NH_3 emissions at leaner conditions mainly stem from reduced burning rate and increased quenching distance.

However, a small fraction of this premixed fuel is not completely oxidized and forms more N_2O than the l-pilot mode forms at a given λ . Almost the entirety of the N_2O produced from dual-fuel NH_3 combustion is fuel-related at higher AES levels [12]. The N_2O emissions decrease under richer in-cylinder conditions because of lower oxygen availability, forming less N_2O predominantly via the $\text{NO} + \text{NH} = \text{N}_2\text{O} + \text{H}$ pathway and, to a lesser degree from the $\text{N}_2 + \text{O} = \text{N}_2\text{O}$ pathway at such high AES levels [12]. For l-pilot, higher in-cylinder temperatures at lower λ values can also promote the decomposition of N_2O that is formed as an intermediate.

To further underscore the fundamental differences in the performance of the two injection strategies (given the AHRRs were qualitatively similar), the indicated N-based emissions are plotted against the ratio of CA5–50 and ignition delay (ID) or Da^* ; see Fig. 5. Note, for simplicity, the ID was computed as the difference between CA5 and the commanded start of diesel injection for the l-pilot cases, whereas the difference between CA5 and start of LTHR was chosen for the E-pilot strategy owing to the presence of LTHR in the bulk premixed phase. The CA5–50 duration has been equated to the main deflagration process in Szybist and Splitter [28]. Da^* serves as a simple qualitative proxy for the ratio of the transport or mixing time scale ($\tau_{\text{transport}}$) to the kinetic time

scale (τ_{kinetic}). The $\tau_{\text{transport}}$ term represents the time required for the unburned reactants to reach the reaction zone or flame front, and τ_{kinetic} refers to the time required for the preignition kinetics (i.e., chemical reactions leading to main heat release). Note that the drawn parallels are a global interpretation of processes occurring locally in-cylinder, and factors such as turbulence complicate the comparison. Furthermore, the relative importance of the transport and kinetic time scales is not being directly compared during the main combustion process here as it would in the true sense of the fundamental nondimensional parameter, Damköhler number (Da). Hence, Da^* should be interpreted with caution and not directly equated to Da.

The results in Fig. 5 suggest that a decrease in λ leads to a considerable reduction in Da^* for l-pilot (from 1.4 at $\lambda = 1.6$ to 0.5 at $\lambda = 1.0$). The l-pilot cases highlight a transition from $\text{Da}^* > 1$ to $\text{Da}^* < 1$ as λ is varied from 1.6 to 1.0. All E-pilot cases, on the other hand, lie entirely in the $\text{Da}^* < 1$ regime. Interestingly, l-pilot cases with a λ of 1.4 have a Da^* of ~ 1 , which also corresponds to an optimal balance between $\text{CO}_2\text{-eq}$ and thermal efficiency benefits, as discussed earlier. Furthermore, the indicated NH_3 and N_2O emissions for l-pilot exhibit a strong linear correlation to Da^* , whereas the trend in indicated NO_x emissions is nonlinear, with a peak at Da^* of 1 ($\lambda = 1.4$). No overall trend emerges for the E-pilot injection strategy between the indicated N-based emissions and Da^* . Note, however, that the AHRR curves for the E-pilot strategy with λ of 1.0 and 1.1, which appear relatively less volumetric in nature and have a partially developed LTHR, align well with the Da^* trends shown for the l-pilot cases. The deviation at the higher λ values may be a result of the modal transition in LTHR behavior. This modal transition appears to affect the formation of N-based species with respect to the defined parameter, Da^* . The presence of LTHR for the E-pilot cases must be considered to explain the differences observed here and necessitates a (0D/1D) kinetic exploration of the influence of mixture formation (post-LTHR) on the bulk gas combustion. Further investigation is also required to understand if and how the observed trends change as a function of AES level for the l-pilot strategy, and whether they hold merit at different engine operating conditions, such as at lower loads and higher engine speeds.

4. Conclusions

In this work, a medium-duty single-cylinder engine was used to investigate the effects of air-fuel equivalence ratio on the combustion performance and emissions of diesel/ NH_3 fuels using l-pilot and E-pilot injection strategies. The experiments were conducted at engine speed and load of 1200 RPM and 12.8 ± 0.2 bar IMEPn (75 % load),

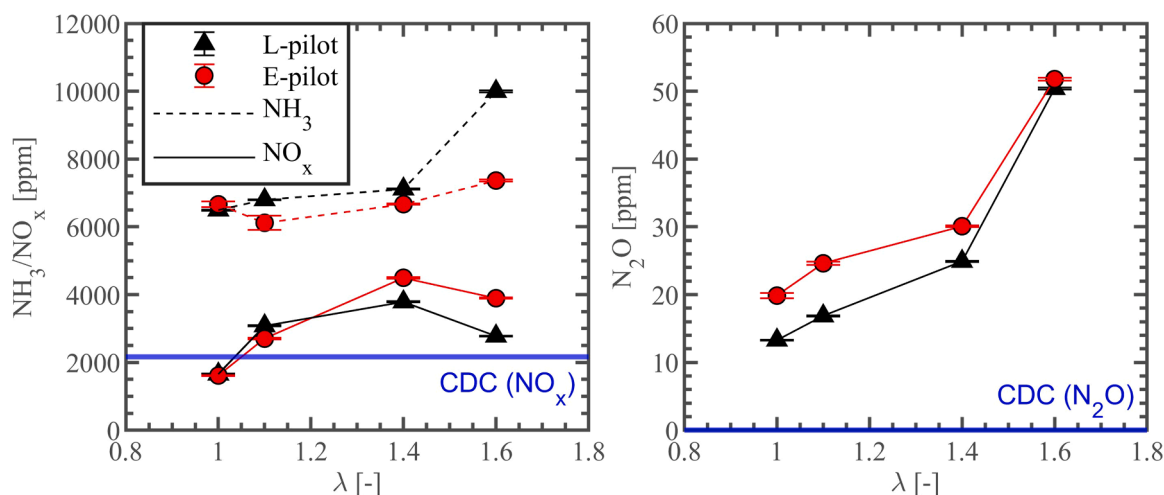


Fig. 4. Engine-out N-based emissions in ppm: NH_3 and NO_x (left) and N_2O (right) as a function of λ . Results for both the l-pilot and E-pilot injection strategies are overlaid. Baseline diesel-only NO_x and N_2O values are also shown for reference. Error bars represent the 95 % confidence interval.

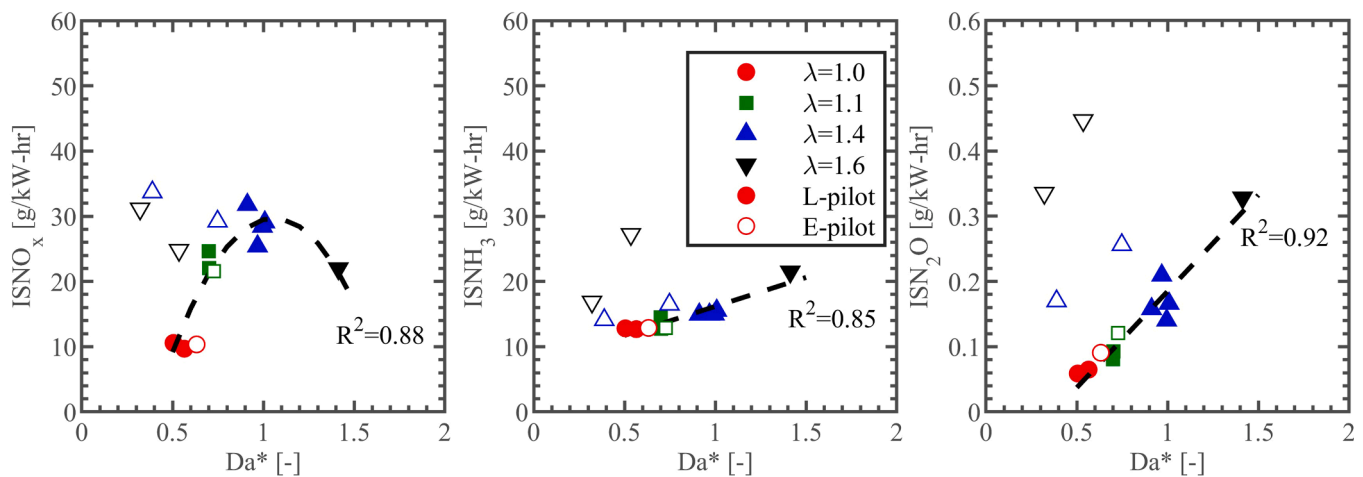


Fig. 5. Indicated specific N-based emissions as a function of Da^* . Note that the l-pilot results are represented by the filled symbols, and E-pilot results are represented by the unshaded symbols. All SOI cases are shown for a given λ value, and the curve-fits are shown for the l-pilot strategy.

respectively. The air flow commanded to the engine was varied to change the global λ with an AES level greater than 90 %. The in-cylinder pressure and AHRR were qualitatively compared for the two injection strategies, and the engine-out N-based emissions were discussed. The CO₂-equivalent emissions and thermal efficiency trade-offs as a function of λ were highlighted, and the ratio of CA5–50 to ID, or Da^* , was used to shed some light on the indicated N-based emissions.

The following conclusions can be drawn from the results of this study:

- AHRR for the richer l-pilot cases appear volumetric in nature with signs of a slow tail at leaner conditions. LTHR is observed for the E-pilot cases, and the combustion process is postulated to be of a lean-premixed reaction propagation type.
- Richer in-cylinder conditions resulted in lower N₂O for both l-pilot and E-pilot modes, mainly driven by lower O₂ availability to form N₂O.
- L-pilot cases with a λ of 1.4 exhibited a Da^* of approximately 1, which also corresponded to an optimal balance between CO_{2,eq} emissions and η_t benefits compared with the baseline CDC case.
- The indicated N-based emissions for l-pilot showed a strong correlation to Da^* , but no apparent trend for E-pilot emerged at the tested boundary conditions.

4.1. Novelty and significance statement

The novelty of this research is that it is the first study (to the best of the authors' knowledge) evaluating the influence of air-fuel equivalence ratio effects and pilot injection strategies on dual-fuel ammonia/diesel combustion at greater than 90 % ammonia energy substitution. It is significant not only because it provides insights into the N-based emissions from a traditional and a non-dimensional perspective, but also because it presents a more fundamental discussion of the heat release rate characteristics of the two injection strategies. Existing gaps in the understanding of dual-fuel ammonia combustion are also concisely highlighted and are of high interest to the global marine engine research and development sector.

CRediT authorship contribution statement

Daanish S. Tyrewala: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis. **Brian C. Kaul:** Writing – review & editing, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Scott J. Curran:**

Writing – review & editing, Resources, Project administration, Funding acquisition, Conceptualization. **Derek A. Splitter:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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